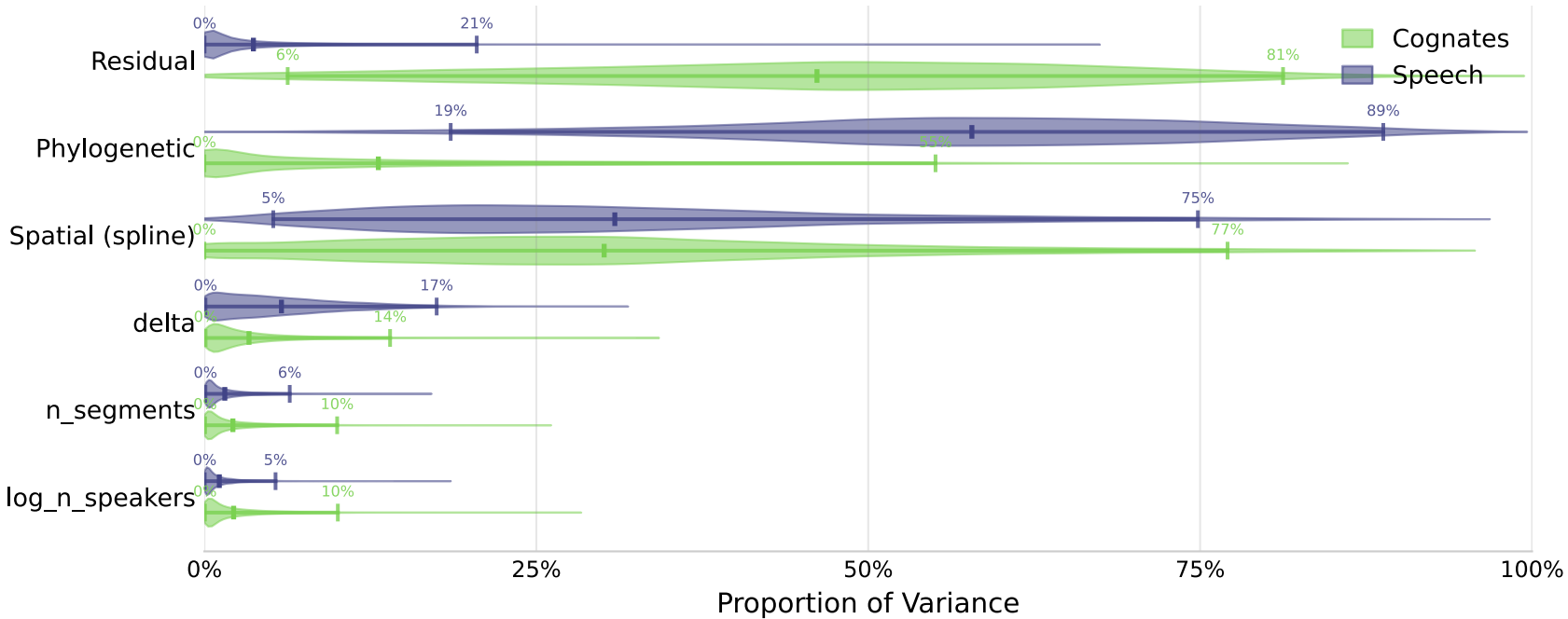
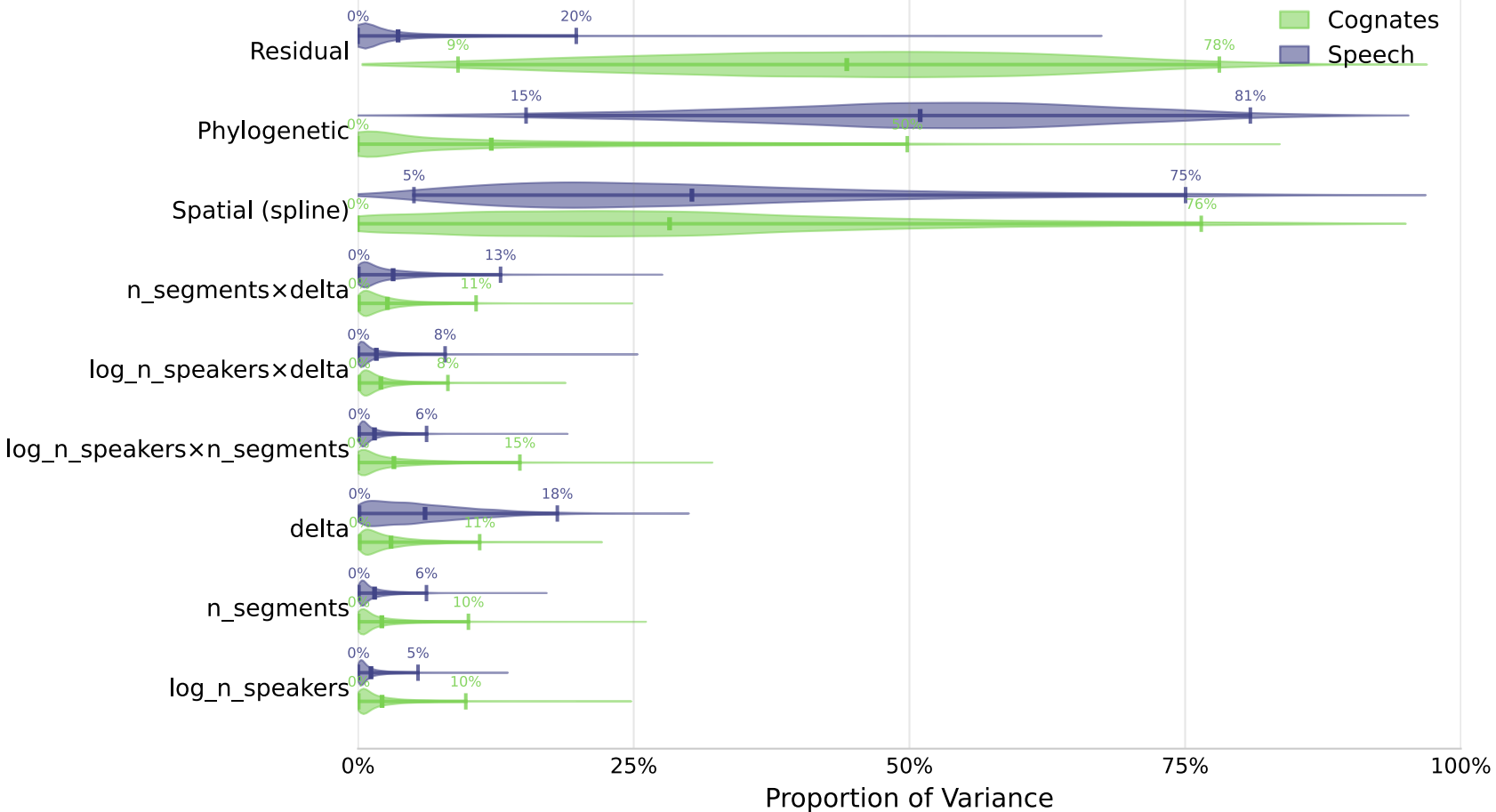




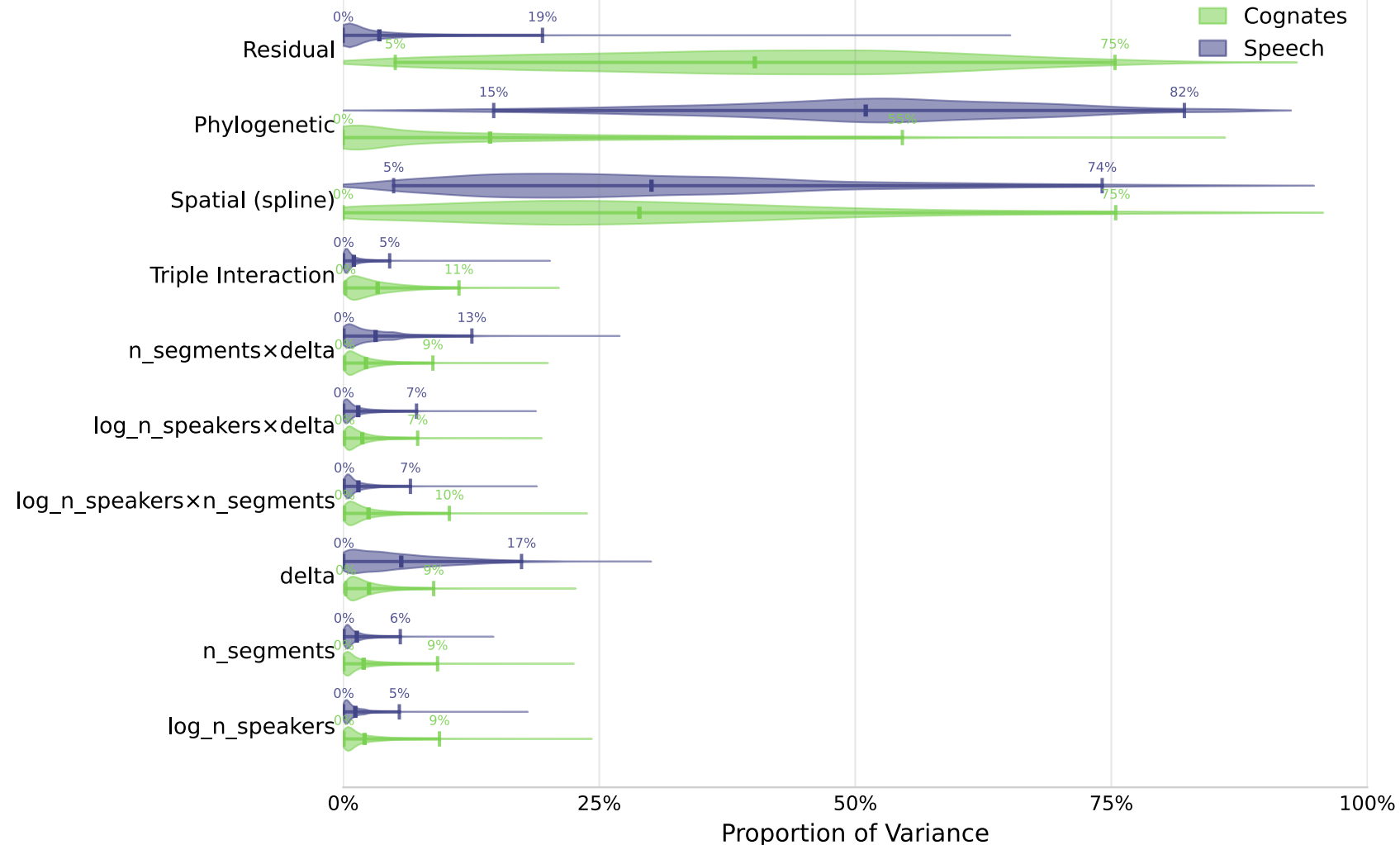
Cognates: ESS_fix=7797/all=7631, $R=1.00$, div=3 | Speech: ESS_fix=5280/all=755, $R=1.01$, div=155

■ Cognates
■ Speech



Speech: ESS_fix=7010/all=634, R=1.01, div=130 | Cognates: ESS_fix=7074/all=6991, R=1.00, div=6

Cognates
Speech



Cognates: ESS_fix=7111/all=5294, R=1.00, div=8 | Speech: ESS_fix=418/all=198, R=1.01, div=515

Coefficient Estimates (95% CI) Main Effects Only

Cognates
Speech

delta

-0.18 0.01

-0.26 0.06

n_segments

-0.06 0.09

-0.18 0.13

log_n_speakers

-0.06 0.08

-0.18 0.12

-0.4

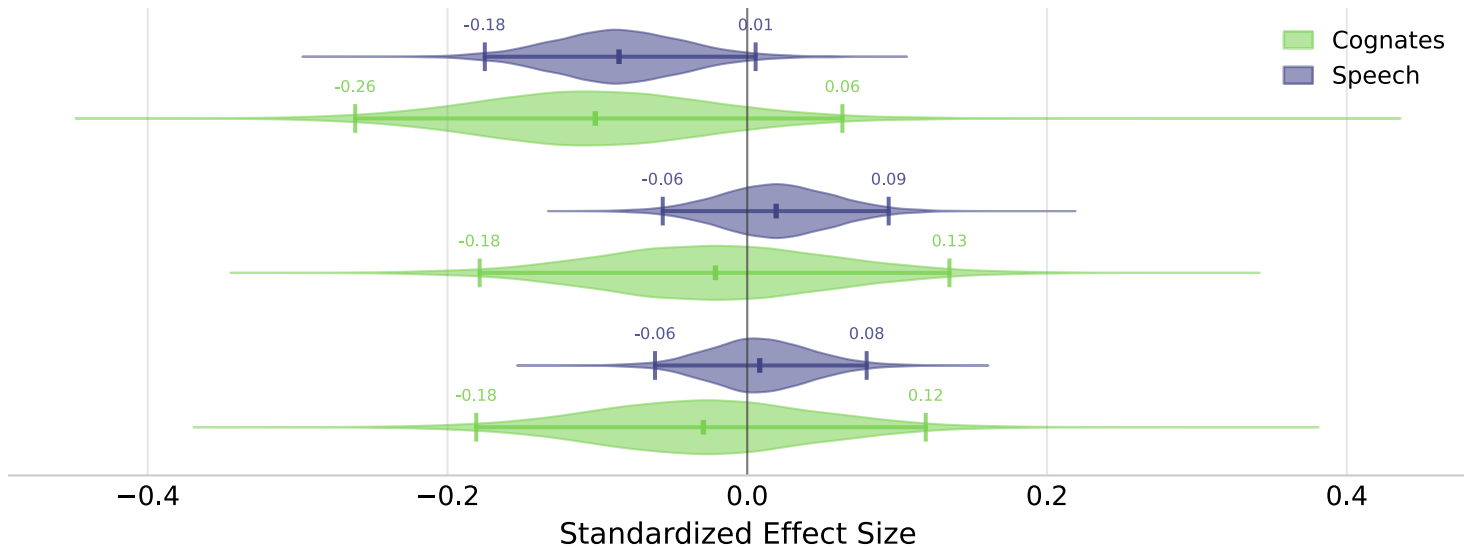
-0.2

0.0

0.2

0.4

Standardized Effect Size



Coefficient Estimates (95% CI) With 2nd-Order Interactions

Cognates
Speech

n_segments×delta

log_n_speakers×delta

log_n_speakers×n_segments

delta

n_segments

log_n_speakers

-1.00

-0.75

-0.50

-0.25

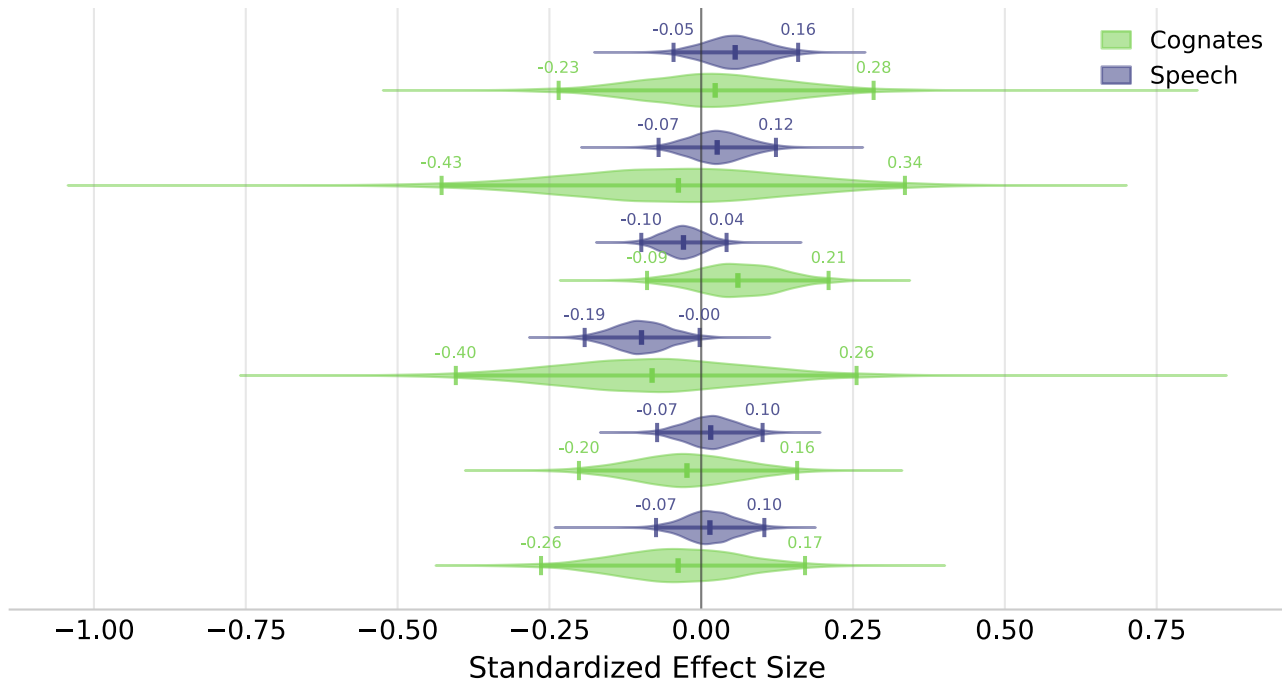
0.00

0.25

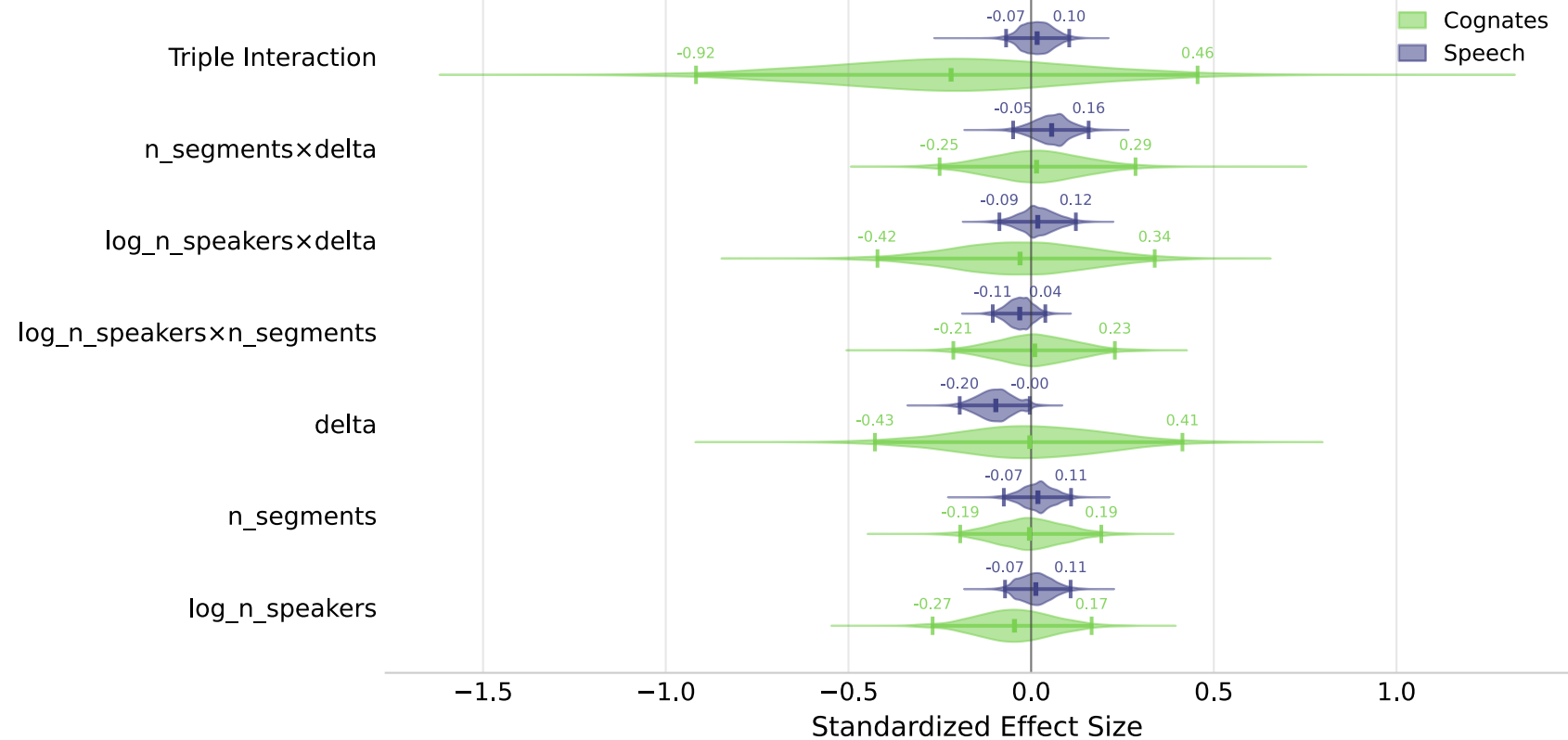
0.50

0.75

Standardized Effect Size

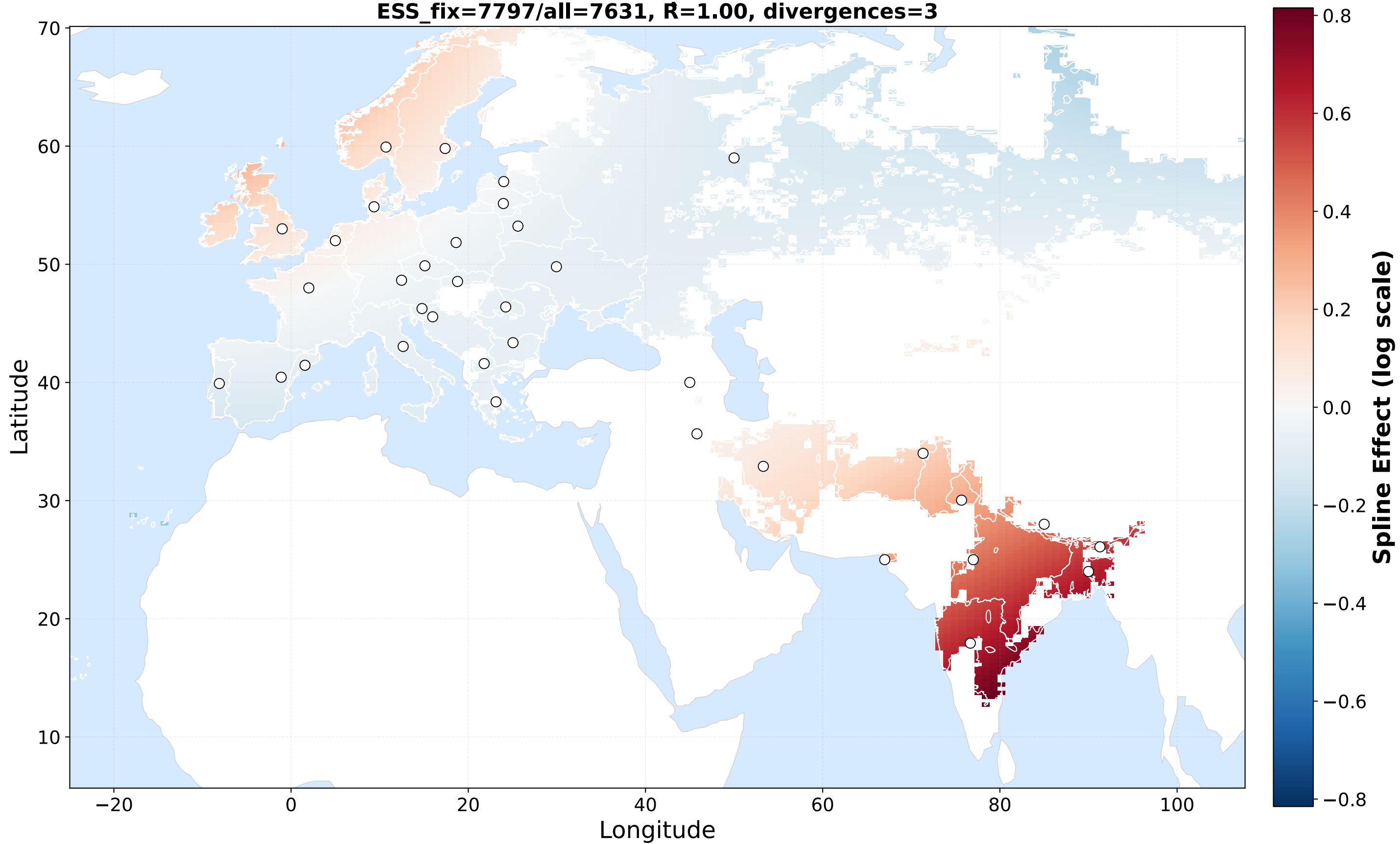


Coefficient Estimates (95% CI)
With All Interactions (2nd + 3rd Order)



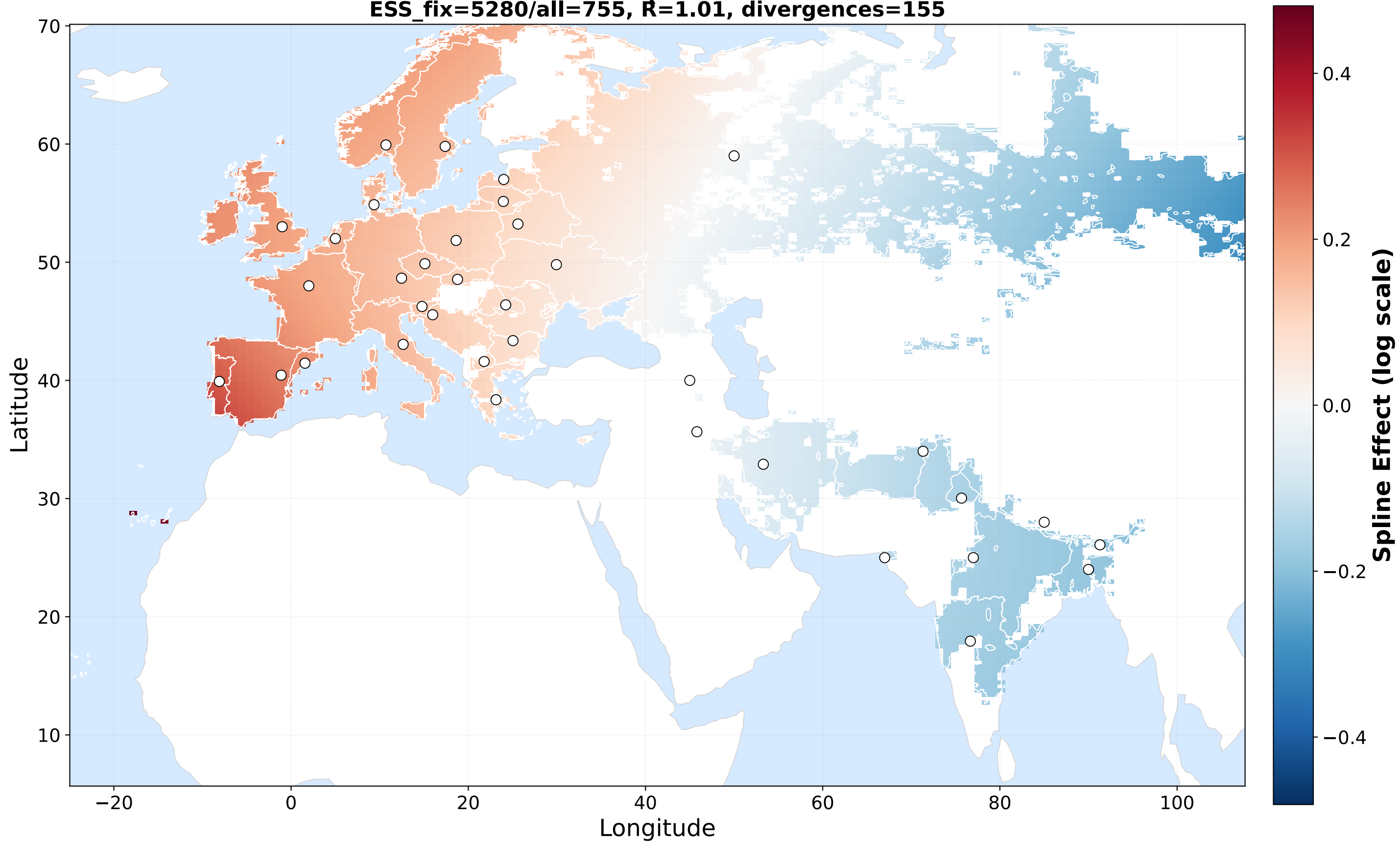
**Spatial Spline Effect: Cognates
(Main Effects Only)**

ESS_fix=7797/all=7631, R=1.00, divergences=3

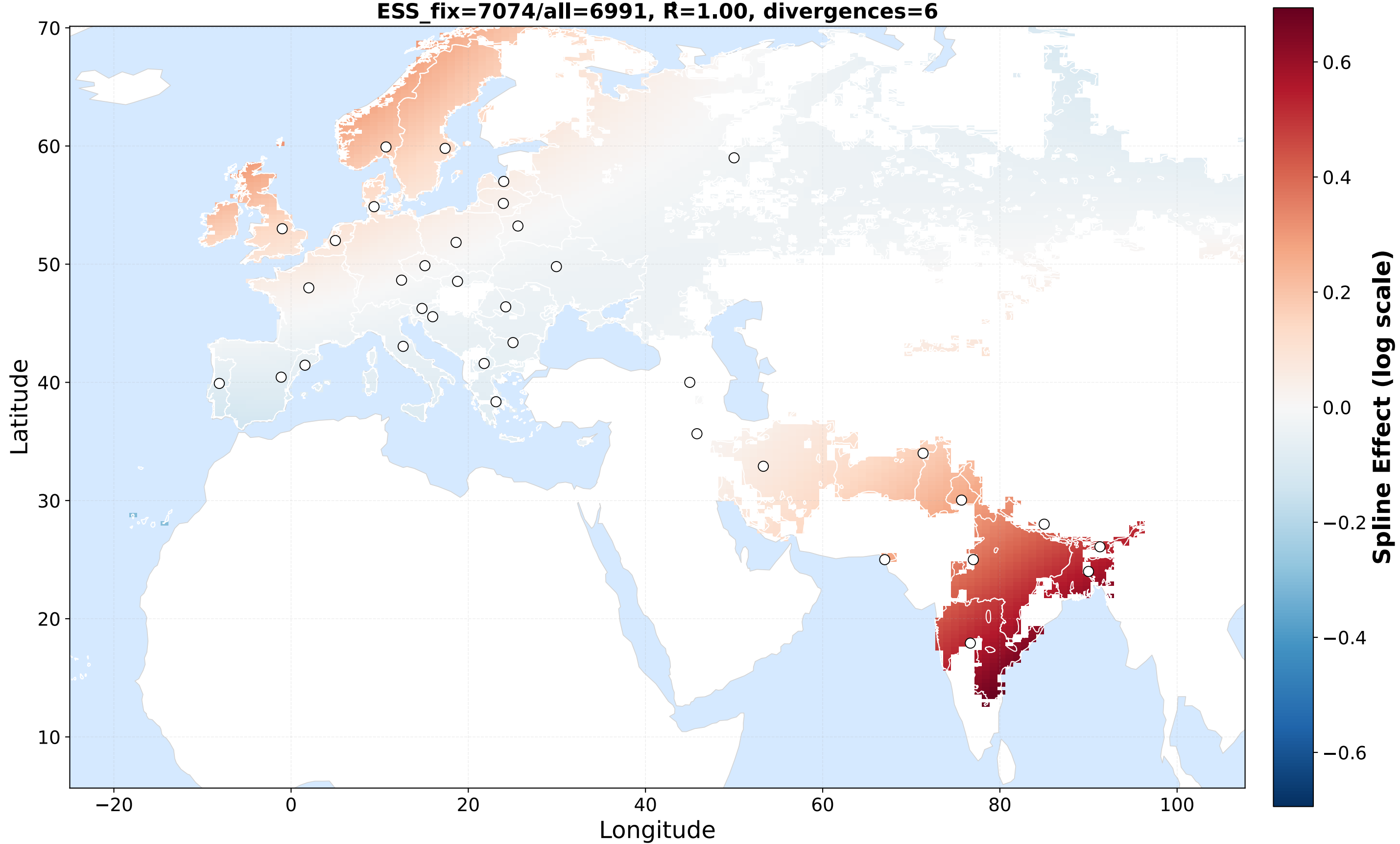


**Spatial Spline Effect: Speech
(Main Effects Only)**

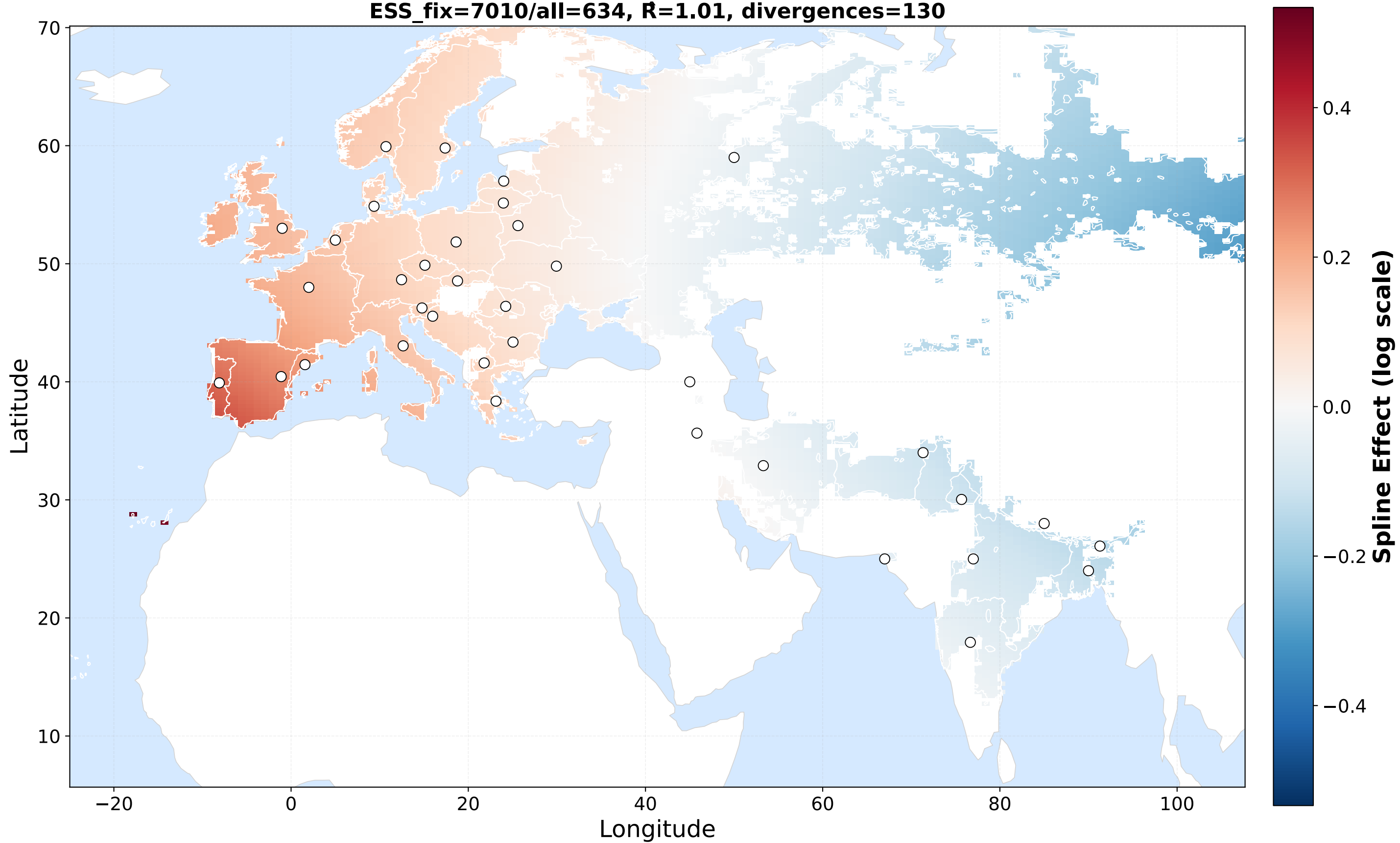
ESS_fix=5280/all=755, R=1.01, divergences=155



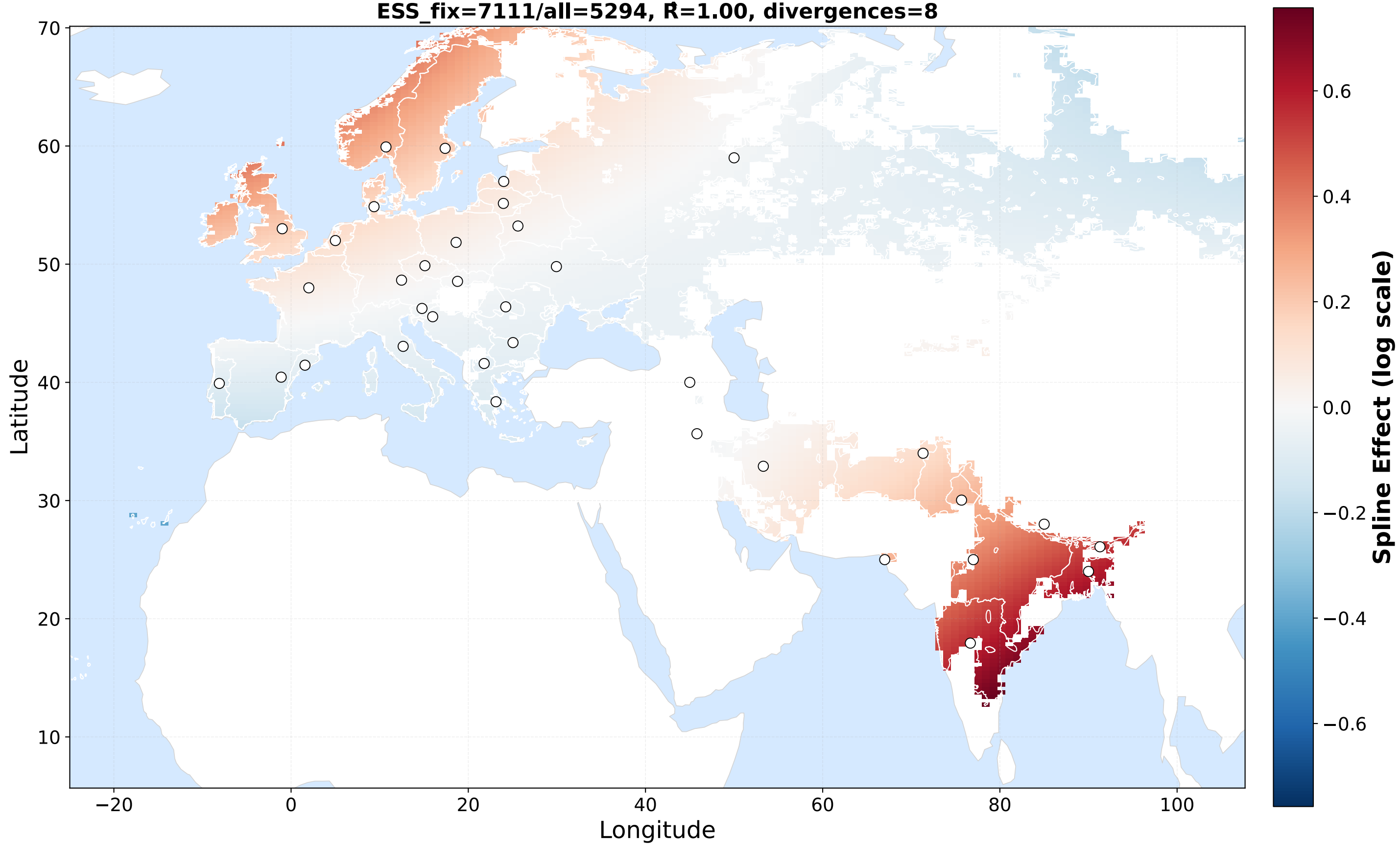
**Spatial Spline Effect: Cognates
(With 2nd-Order Interactions)**
ESS_fix=7074/all=6991, R=1.00, divergences=6



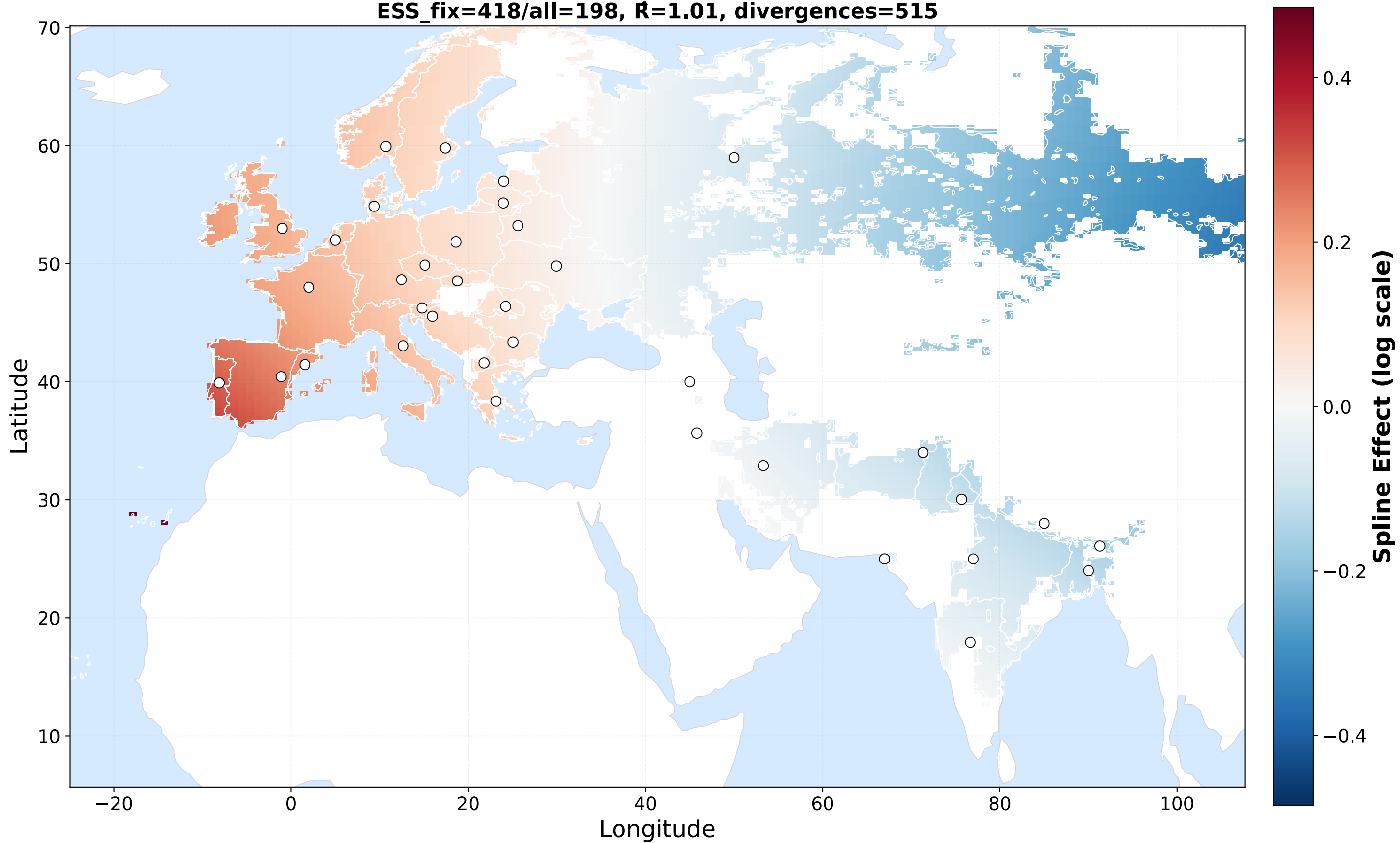
Spatial Spline Effect: Speech
(With 2nd-Order Interactions)
ESS_fix=7010/all=634, R=1.01, divergences=130



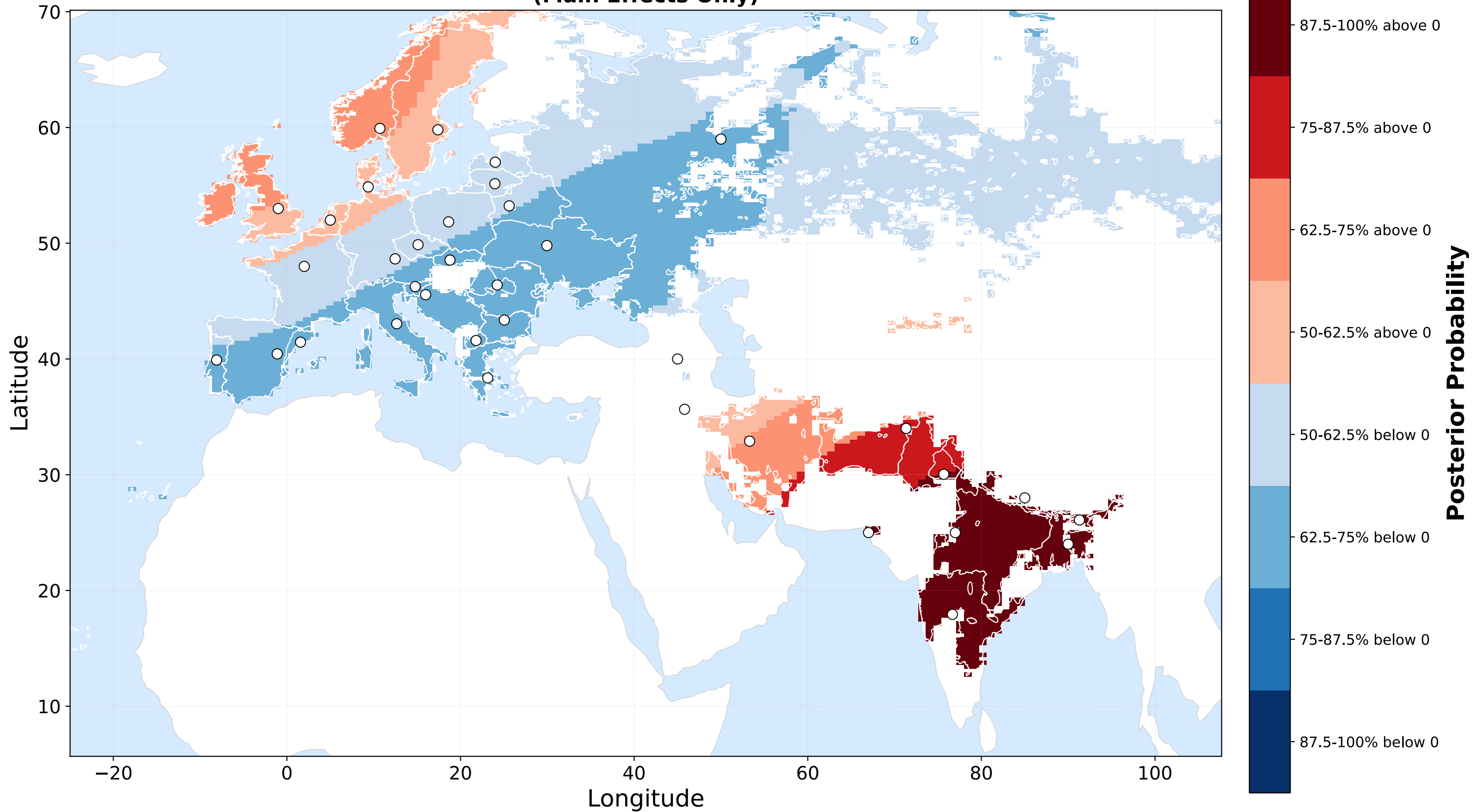
Spatial Spline Effect: Cognates
(With All Interactions (2nd + 3rd Order))
ESS_fix=7111/all=5294, R=1.00, divergences=8



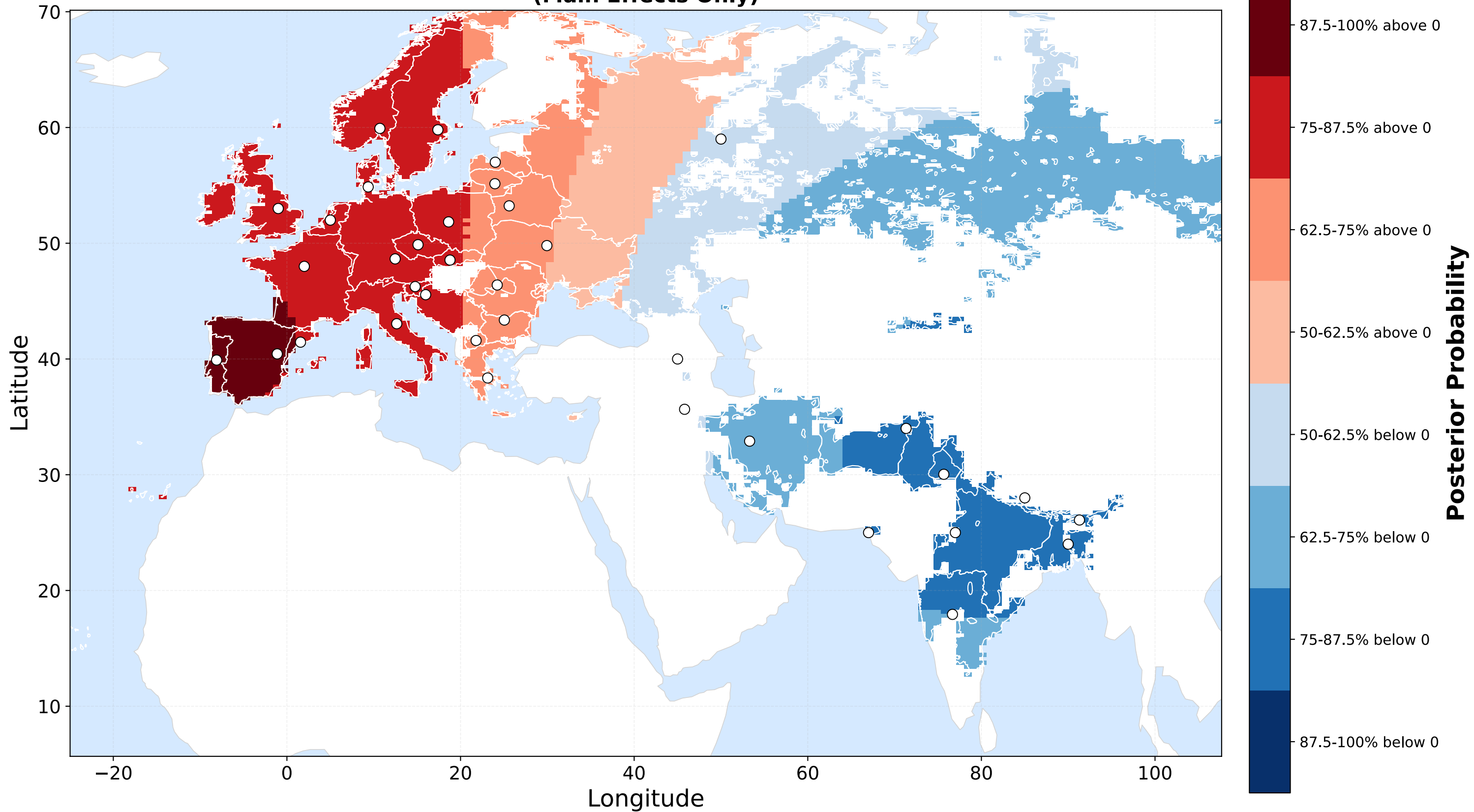
Spatial Spline Effect: Speech
(With All Interactions (2nd + 3rd Order))
ESS_fix=418/all=198, R=1.01, divergences=515



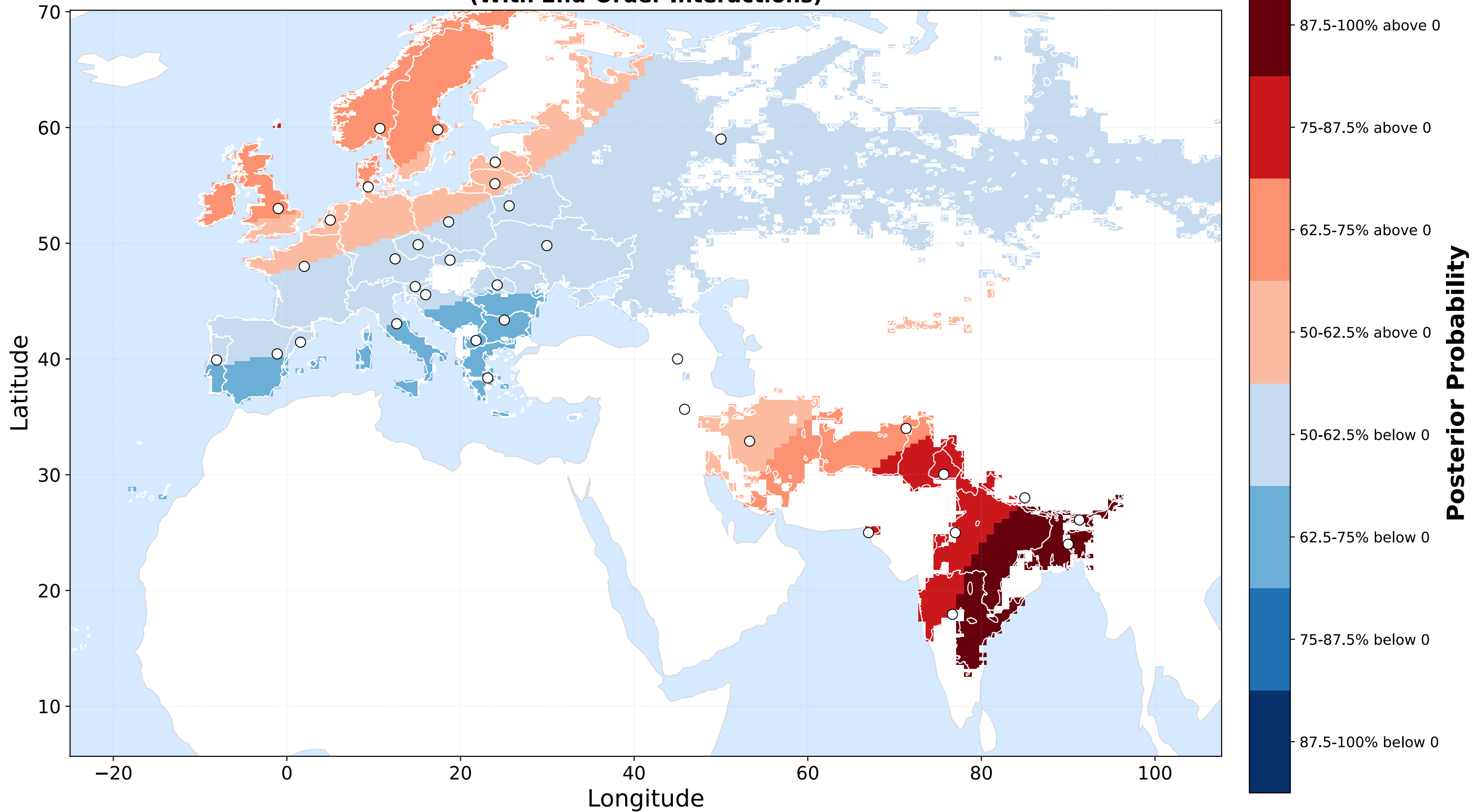
**Posterior Probability of Positive Spline Effect: Cognates
(Main Effects Only)**



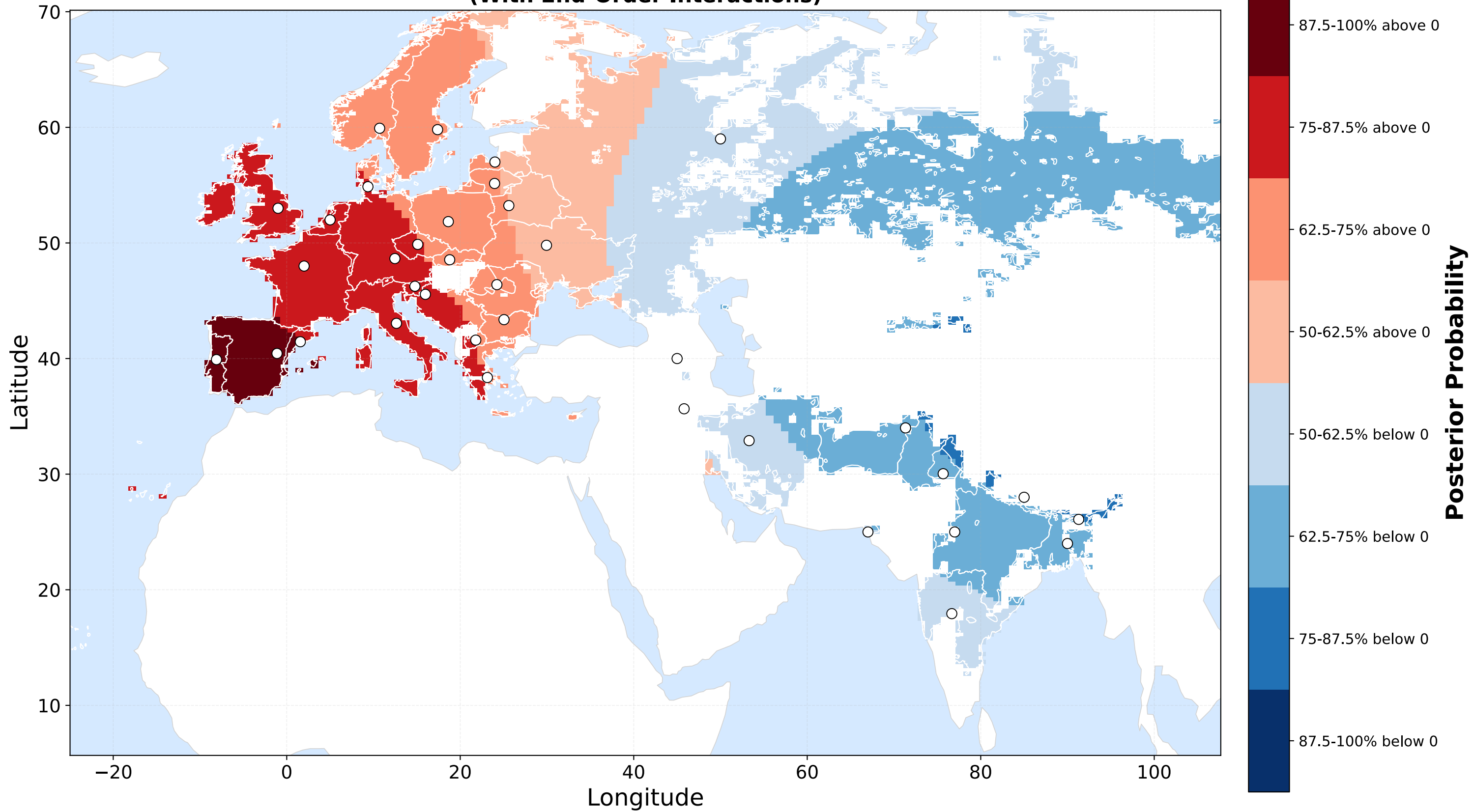
**Posterior Probability of Positive Spline Effect: Speech
(Main Effects Only)**



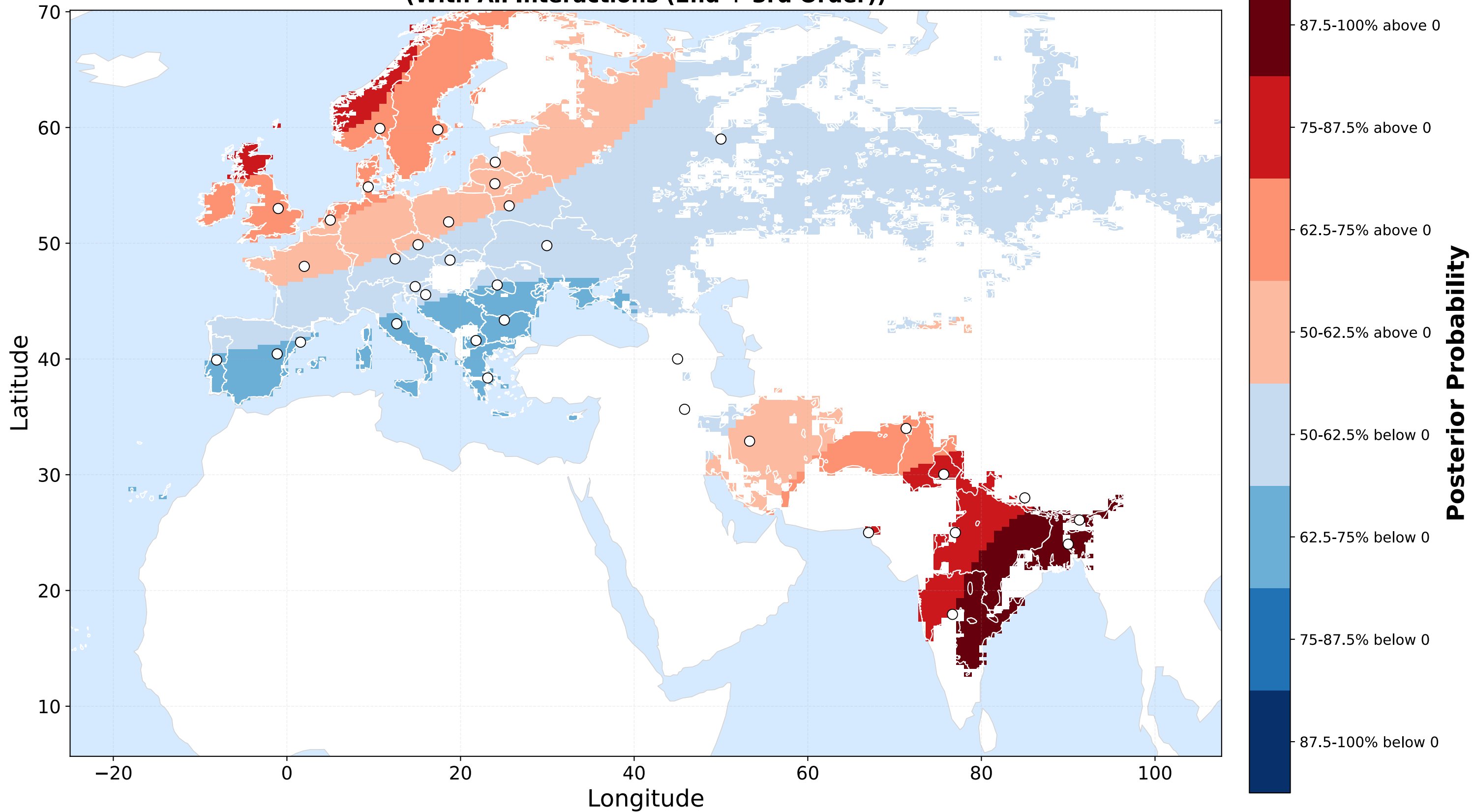
**Posterior Probability of Positive Spline Effect: Cognates
(With 2nd-Order Interactions)**



**Posterior Probability of Positive Spline Effect: Speech
(With 2nd-Order Interactions)**



**Posterior Probability of Positive Spline Effect: Cognates
(With All Interactions (2nd + 3rd Order))**



**Posterior Probability of Positive Spline Effect: Speech
(With All Interactions (2nd + 3rd Order))**

